

MEDA: MoE-based Concept Drift Adaptation for In-vehicle Network Intrusion Detection

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Abstract—Recent years have witnessed the increasing popularity of Internet of Vehicles (IoV), meanwhile the number of cyber threats have surged significantly. Various approaches have been previously proposed to mitigate the potential cyber threats within in-vehicle network. However, the concept drifts derived from ever-evolving offensive tactics bring challenges to the pre-built detection system, resulting in the degradation of detection performance under real-world setting. In this paper, we propose a concept drift adaptation framework MEDA based on Mixture of Experts (MoE) method for constructing a real-time self-updating intrusion detection system on in-vehicle network data. We use the combinations of several drift detection and adaption methods as individual experts, and leverage a gating function to determine the probability and weights of adopting each expert module. Experiments conducted on two publicly available datasets demonstrate the effectiveness of our proposed method in comparison to state-of-the-art approaches.

Index Terms—mixture of experts, in-vehicle network, intrusion detection, network security, machine learning

I. INTRODUCTION

With the rapid development of modern vehicle technology, such as self-driving automobile and intelligent connected vehicles, the Internet of Vehicles (IoV) has emerged as a crucial component of current ubiquitous connectivity [1]. IoV systems typically consist of intra-vehicle networks (IVNs) and external networks. In IVNs, the Controller Area Network (CAN) bus is the core infrastructure enabling communication between Electronic Control Units (ECUs) to implement versatile functionalities, thereby gathering massive sensor data and offering various control capabilities [2]. Unlike the widespread Ethernet network, control signals are broadcast in the form of several short messages that describe the vehicle status and convey user instructions via the CAN. In result, the

lack of security attributes during the design of the CAN bus communication protocol leads in-vehicle networks to potential threats such as data spoofing or denial-of-service attacks [3], causing severe risks to both property and life safety in physical world.

To address this cyber-security problem, monitoring real-time communication traffic data within the CAN bus and alerting potential cyber threats is considered as a viable solution [4] [5]. Through the investigation upon the network traffic data within the Controller Area Network (CAN) bus, typical attack behaviors and anomalous traffic patterns can be abstracted, thereby facilitating the detection of ongoing cyber intrusion threats. Presently, the primary methodologies can be categorized into signature-based intrusion detection systems (IDS) [6], [7] and machine learning-based IDS [2]. Although signature-based methods require less computation time and can be applied to resource constrained invehicle ECUs, they cannot recognize potential unseen threats and . In contrast, machine learning-based IDS relies on architectures such as Random Forests [3], [8], [9] and Neural Networks [10]–[12] to perform automatic feature extraction and statistical analysis of incoming network traffic data. Next, they utilize optimization metrics, including distance or similarity measures, and employs intricate neural computations to discern the abnormality of the input traffic compared to benign traffic, thereby determining the presence or absence of network threats [13]. Besides, due to the generalization and extrapolation capabilities of machine learning technology, machine learning-based methods have the potential to identify unknown intrusion threats to a certain extent, making it a promising direction for future research and development.

However, most existing intrusion detection methods for in-vehicle network rely heavily on pre-built threat signatures or seen CAN data used in training phase, making it challenging to defending against emerging threats derived from concept

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drifts during practical applications. That means continual and promptly updates or patches are needed along with the utility of in-vehicle IDS, which leads to the potential damage caused by emerging threats during the intervals between such updates. Consequently, the capability of autonomous upgrading and adaptive to ever-evolving CAN communication traffic patterns has become a significant research focus aimed at solving real-world application challenges of in-vehicle IDS.

In this paper, we propose a vehicle-agnostic concept drift adaptation framework namely MEDA, for improving the in-vehicle network intrusion detection performance on the drift data on the basis of Mixture-of-Experts architecture [14]. We integrate several commonly-used drift detection and adaption methods, consisting of adaptive windowing (ADWIN) [15], drift detection method (DDM) [16], Kolmogorov-Smirnov Windowing (KSWIN) [17], adaptive random forest (ARF) [18] and streaming random patches (SRP) [19], to construct base experts. And we design a gating function to determine select which expert to predict when concept drift occurs.

The main contributions of this paper are as follows:

- It designs a MoE-based concept drift adaptation framework MEDA.
- It illustrates the potential concept drift problems under the in-vehicle network scenarios.
- It evaluates the proposed framework on two public in-vehicle network traffic datasets, including CarHack 2009 [10], [11] and CICIoV 2024 [20], in comparison with other commonly-used approach.

The structure of this paper is organized as follows. Section II introduces the preliminary. Section III elaborates the proposed MoE-based concept drift adaption framework in this paper. And Section IV presents and discusses the experimental results. Finally, Section V concludes the paper.

II. PRELIMINARY

A. CAN-based In-vehicle Network and Cyber Threats

In order to capture data from vehicle sensors and transmit intricate operational commands to drivers, ECUs (Electronic Control Units) distributed across powertrains, chassis control systems, body control systems, and in-vehicle entertainment systems, are linked through a common non-authenticated broadcast channel known as CAN bus. A practical skeleton of the modern vehicular ecosystem is illustrated in Fig. 1 as follows:

Communication data on the CAN bus is segmented into CAN frames, as illustrated in Fig. 2. Within each frame, the ID field (comprising the first 11 bits of the arbitration field) serves to identify the originator of the traffic data, while the data field (comprising 8 bytes of the data segment) specifically denotes the content of the transmitted data.

The numerical value of the CAN bus ID dictates the priority of message processing. In principle, message senders on the CAN bus occupy unique IDs for transmitting messages. However, in practical scenarios, the transmission of CAN frames in an unauthenticated broadcast format enables attackers to

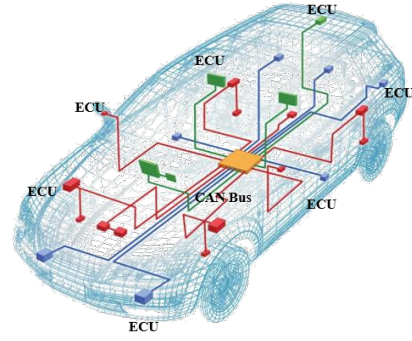


Fig. 1. In-vehicle communication of ECUs over CAN Bus

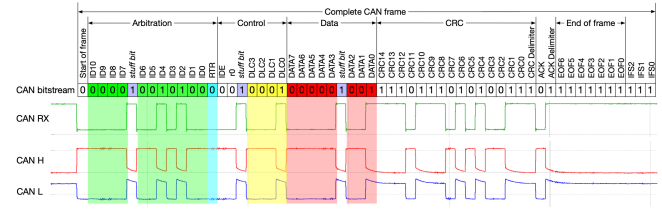


Fig. 2. The structure of a common CAN Bus frame

infiltrate the CAN bus and impersonate in-vehicle ECUs by forging CAN IDs, thereby causing a potential threat to automotive security, such as:

- DoS Attack: Injecting messages of '0000' CAN ID. Since '0000' is the most dominant, the injected messages will occupy the foremost position in the message processing sequence, thus obstructing the processing of normal messages.
- Fuzzy Attack : Injecting messages of totally random CAN ID and DATA values as junk messages.
- Spoofing Attack (RPM/gear) : Injecting messages of certain CAN ID related to RPM/gear information , engaging in deception against the system.

B. Concept Drift and Concept Adaption

Given the dynamic and non-stationary nature of in-vehicle network environments, the analysis of in-vehicle network data frequently confronts challenges stemming from concept drift, where the underlying data distributions vary over time [21].

$$f_t(X; y) \neq f_{t+1}(X; y) \quad (1)$$

The emergence of concept drift phenomena often undermines the performance of deployed intrusion detection models, incurring security concerns. As shown in (??), when concept drift occurs, the feature distribution changes. This implies that the previously established mapping relationships $X \Rightarrow y$ constructed by the trained model will turn invalid. Concept drifts can be categorized into two primary types: sudden and gradual, based on the rate of change in data distributions. To effectively cope with concept drift, an intrusion detection model must be of capable of precisely identifying the occurrence of drifts and promptly adapt its parameters to accommodate these changes,

thereby maintaining a high level of prediction accuracy and resilience against evolving data patterns.

C. Mixture-of-Experts System

The Mixture-of-Experts (MoE) architecture refers to a system governed by a supervisory process that comprises multiple distinct modules, termed ‘experts,’ each tasked with processing a portion of the input data. Each individual ‘expert’ exhibits proficiency in specific regions of the input space, and during the determination of the output, the result provided by the most competent expert is selected as the output of the entire MoE system.

Since distinct concept drift detection and adaptation approaches exhibit varying performance across different incoming data. In the context of our work, we draw inspiration from current Mixture of Experts (MoE) paradigm. By integrating multiple distinct base drift detectors, we aim to achieve an enhanced capability for drift detection and adaptation.

III. METHODOLOGY

A. Overview of Proposed Framework

Fig. 3 offers a comprehensive depiction of the MEDA framework. The core procedures of the proposed framework are enumerated below. Initially, the continuously incoming CAN frame streams undergoes sampling through the application of the sliding-window sampling technique, resulting in a condensed yet highly representative subset.

Subsequently, distinct concept drift adaptation methodologies are devised and established as foundational experts, tasked with both initial anomaly detection and subsequent drift adaptation. Then a gating function is formulated, strategically integrating the selection probabilities derived from these four base experts, grounded within the proposed MEDA framework. Ultimately, by combining insights from individual expert modules, a comprehensive decision can be made to adapt to concept drifts, ensuring an automatic drift adaptive in-vehicle intrusion detection system.

B. Data Pre-processing and Sampling

As CAN frame streams are continually-generated along with the runtime of automotive systems, usually producing vast amounts of data within a short period of time, which is infeasible and resource-consuming for analysis. Therefore, the implementation of an efficient data sampling strategy is crucial for selecting highly representative data samples while discarding unnecessary and redundant information. In our proposed system, we employ the “sliding window” sampling approach to extract a representative subset of the data. This method efficiently groups the data samples into distinct clusters, enabling the selection of a proportional number of samples from each window, where samples within the same cluster exhibit similar characteristics.

Given the substantial size of in-vehicle CAN bus traffic datasets, we leverage the “sliding window” method to select a manageable 5% of the original data samples for evaluating our

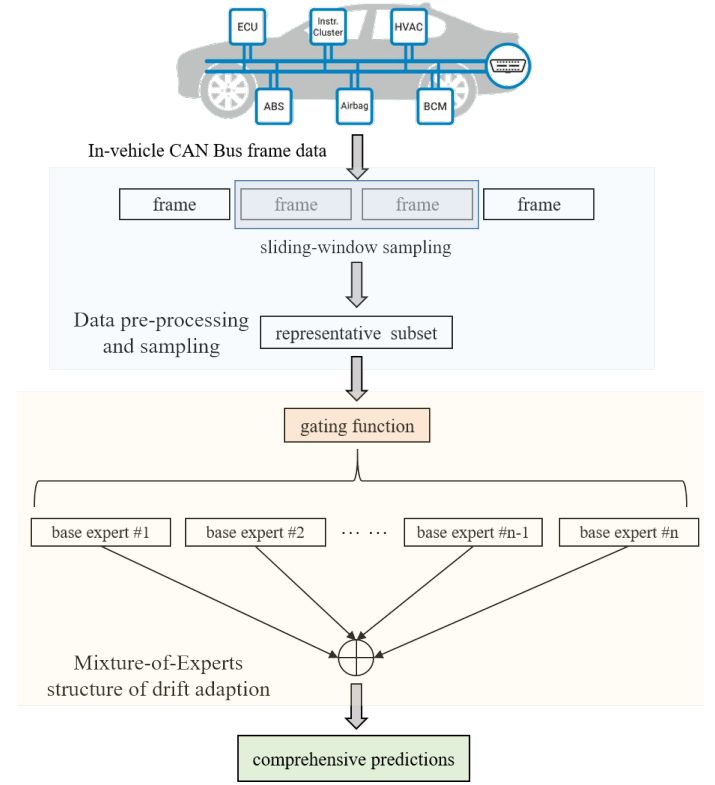


Fig. 3. The proposed drift adaptive in-vehicle intrusion detection framework

proposed framework. The resultant subset size is flexible, contingent upon factors such as the rate of CAN data generation and the computational capabilities of the server infrastructure. This approach ensures that our analysis remains both feasible and informative, focusing on the most pertinent data points.

C. Concept Drift Adaptation from Mixture-of-Experts

We choose adaptive windowing (ADWIN), drift detection method (DDM), Kolmogorov-Smirnov Windowing (KSWIN) as three basic drift detection algorithms, combined with two state-of-the-art drift adaptation methods (adaptive random forest (ARF) & streaming random patches (SRP)) to build base expert. Thus, we can get six base experts, namely ARF-ADWIN, ARF-DDM, ARF-KSWIN, SRP-ADWIN, SRP-DDM and SRP-KSWIN.

In our design, DDM exhibits commendable performance in detecting sudden drift scenarios, whereas ADWIN has a stronger capability in identifying gradual drift patterns. Furthermore, KSWIN stands out with its superior ability to detect both these types of drift effectively. Integrating such drift detection methods enables the proposed MoE model to detect both drift types effectively. Once we get drift alert from the drift detection component such as ADWIN, DDM or KSWIN, we will subsequently facilitates the adaptation of our model to the new conditions, ensuring continued relevance and accuracy.

As the back-end adaption module, ARF and SRF adapt the original model to fit new-coming data. Both ARF and SRF integrate random forest algorithms with online learning capabilities, enabling it to dynamically incorporate additional decision trees in response to changes within data streams. In details, ARF relies on local subspace randomization and SRP uses a global subspace randomization, which signifies that these two approaches can complement each other's strengths. After the calculation, each individual expert outputs more accurate detection results for the selection of MoE module.

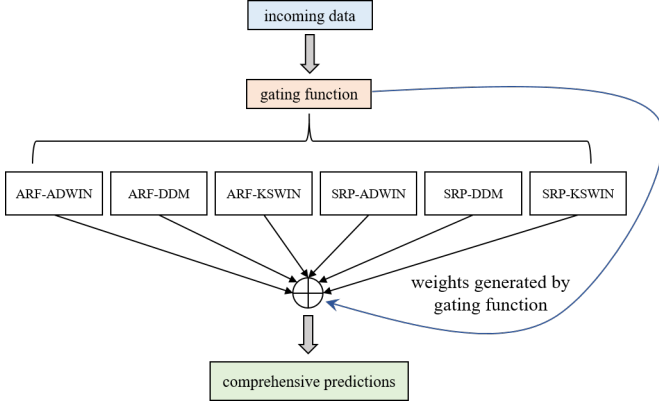


Fig. 4. The detailed MoE structure in proposed framework

As illustrated in Fig. 4, the gating function $G(x)$ works as a core component to select some certain experts.

Unlike the MoE module used in popular large language model, the gating function in our framework relies on the loss of each expert rather a trainable neural network, as illustrate in (2) and (3).

$$G(x) = \text{Softmax}(\text{KeepTopK}(H(x), k)) \quad (2)$$

k stands for the top k experts will be activated for calculation. In our framework, k is set as 3.

$$H(x) = 1/\text{loss}(E_i(x)) \quad (3)$$

Subsequently, we generate weights for the calculation of eventual output as (4). err stands for the err_proba of current prediction.

$$y = \sum_{i=1}^n G(x)W_i(x)E_i(x) \quad (4)$$

While the weight can be calculated as the (5).

$$W_i(x) = \frac{1/(1 - \text{err}_i)}{\sum_{j \in \text{Topk}} 1/(1 - \text{err}_j)} \quad (5)$$

IV. EXPERIMENTAL EVALUATION

A. Experiment Setting

In this section, we introduce the evaluation of proposed methodology on two public datasets. All experiments in this paper are conducted on a laptop with a CPU of Intel i9 12900H, RAM 64 GB 4,800 MHz, and a GPU of NVIDIA GeForce RTX 3060 6GB.

B. Datasets

Two datasets are used to evaluate the proposed framework.

The first dataset is the CarHack dataset [10], [11] constructed by logging CAN traffic via the OBD-II port from a real vehicle while message injection attacks were performing. The dataset contains each 300 intrusions of message injection in total, including DoS attack, fuzzy attack, spoofing the drive gear, and spoofing the RPM gauge.

The second dataset is CICIoV 2024 [20], published by University of New Brunswick, as a realistic benchmark dataset. CICIoV 2024 contains five attacks against the fully intact inner structure of a 2019 Ford car, including DoS, spoofing-GAS, spoofing-RPM, spoofing-SPEED, and spoofing-STEERING_WHEEL, complete with all Electronic Control Units (ECUs).

We focus on some part of above two datasets, where attack patterns change over time, causing multiple concept drifts in the both two datasets.

C. Experiments

In our experiment, we streamlined the labeling process by unifying various types of network threats within the dataset under the simplified category of “malicious”. Consequently, the in-vehicle network intrusion detection problem was de facto transformed to a binary classification task.

We conducted experiments on CarHack dataset and CICIoV 2024 dataset respectively, in comparison with state-of-the-art drift adaptive approaches including HT [22], LB [23], and six base experts within the MoE.

Table I and Table II show the performance comparison of the proposed MEDA framework with other methods on both two datasets. As shown in Table I, the proposed MEDA framework outperforms all other approaches in average accuracy, average f1-score, lowest accuracy, lowest f1-score. In addition, the proposed MEDA framework consume the shortest time to adapt to the new-coming data (the amount of instances for adaptive learning to achieve the 90% accuracy before concept drift).

The experimental results demonstrate that our proposed MEDA framework is capable of promptly self-updating under the ever-evolving data environment, thereby maintaining a high detection performance to counter various network threats targeting in-vehicle networks. Meanwhile, the incorporation of the MoE architecture enables the complementary strengths of individual experts to be harnessed, resulting in an enhanced data adaptation capability that surpasses the capabilities of any single expert.

V. CONCLUSION

In this paper, we propose a drift adaptive in-vehicle network intrusion detection framework, named MEDA, based on the MoE combination of six state-of-the-art drift adaptation methods. Experiments show our proposed MEDA can achieve a better performance than previous approaches, and eliminates the duration of response to emerging cyber threats. In future

TABLE I
COMPARATIVE EVALUATION ON CARHACK DATASET

Method	Average Accuracy(%)	Average F1-Score(%)	Lowest Accuracy(%)	Lowest F1-Score(%)	Average Adaption Time(Instances.)
HT	96.72	95.88	91.07	92.01	197
LB	96.83	96.04	92.09	93.14	223
ARF-ADWIN	97.88	98.06	93.29	94.01	166
ARF-KSWIN	98.25	98.40	94.35	94.71	150
ARF-DDM	98.22	98.37	93.31	93.01	175
SRP-ADWIN	98.57	98.56	94.20	94.29	141
SRP-KSWIN	98.77	98.69	94.42	94.85	130
SRP-DDM	98.91	99.02	93.57	93.40	153
Proposed MEDA	99.01	99.04	94.83	95.09	101

TABLE II
COMPARATIVE EVALUATION ON CIC-IoV 2024 DATASET

Method	Average Accuracy(%)	Average F1-Score(%)	Lowest Accuracy(%)	Lowest F1-Score(%)	Average Adaption Time(Instances.)
HT	97.43	96.52	92.11	93.06	236
LB	98.12	98.07	93.20	94.35	264
ARF-ADWIN	98.27	98.54	93.56	94.28	184
ARF-KSWIN	98.79	99.02	94.01	94.50	170
ARF-DDM	98.53	98.91	92.41	93.17	192
SRP-ADWIN	98.73	98.84	94.27	94.35	161
SRP-KSWIN	99.03	99.11	94.69	95.05	137
SRP-DDM	98.91	99.02	93.57	93.40	169
Proposed MEDA	99.19	99.21	95.43	95.79	119

work, the proposed framework can be enhanced with more SOTA base expert for better performance and diversity.

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